THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 15	Squares from 15^2 to 29^2
2	**(c)** 2^8	$2 \times 2^7 = 2^8$
3	**(d)** 5	$3b + 4 = 19$
4	**(b)** 90°	One part $= 18^\circ$
5	**(a)** 41	LCM plus the common remainder
6	**(c)** 53	$x^2 + y^2 = (x + y)^2 - 2xy$
7	**(c)** 8 : 12 : 21	Make the middle terms equal using their LCM
8	**(d)** 1%	Loss % = (p²/100) = 10²/100
9	**(c)** 120	Legs 20 and 48 since $5^2 + 12^2 = 13^2$
10	**(d)** 24	Let ages be $3x$, $5x$: $(3x + 8) : (5x + 8) = 2 : 3$ gives $x = 8$
11	**(c)** 20	$E = 5F/2 = 30$; $V = 2 + E - F$
12	**(d)** 42	$5 \times 30 - 4 \times 27$
13	**(b)** 2	$9(a - b) = 9d$
14	**(d)** 768 cm²	$l + b = 64$, $l = 3b$: $b = 16$, $l = 48$
15	**(d)** 62	Next square is $36^2 = 1296$
16	**(b)** 4	$2x + 1 = 9$
17	**(d)** 5	Number of binary digits of the largest weight to be measured
18	**(c)** 60°	$x + 2x = 180$
19	**(c)** 57	71 multiples up to 500 minus 14 multiples below 100
20	**(a)** 49	$x^2 + y^2 = (x - y)^2 + 2xy$
21	**(c)** 24 years	Let the ages be $3x$ and $2x$; solving $(3x + 4) \times 5 = (2x + 4) \times 7$ gives $x = 8$
22	**(b)** 60500	50000×(1+10/100)²
23	**(c)** 972	Sides 27 and 36; area $= 27 \times 36$
24	**(b)** 1350	One part $=$ ₹450
25	**(b)** 8	$V = 2E/3 = 12$; $F = 2 + E - V$
26	**(c)** 41	Correct sum $= 360 - 30 + 80$
27	**(b)** 7	$(x + 1/x)^2 = x^2 + 2 + 1/x^2$
28	**(b)** 128 cm²	Height $= \sqrt{10^2 - 6^2} = 8$ cm; area $= \frac{1}{2}(10 + 22) \times 8$
29	**(b)** 77	Previous square is $50^2 = 2500$
30	**(b)** 7	$2^4 \times 5^7 = 10^4 \times 125$
31	**(b)** 32_5	Convert every number to decimal and compare
32	**(a)** 130°	A diagonal bisects the vertex angles; $\angle BCD = 2x$ and $\angle B$ is supplementary to it
33	**(a)** 112	Count multiples from 100 to 999
34	**(b)** 24000	$(512 + 488)(512 - 488)$
35	**(c)** ₹ 600	Convert the reciprocals to whole-number ratio using the LCM
36	**(a)** 40500	50000×(1−10/100)²
37	**(c)** 15, 112, 113	Use $(n, \frac{n^2-1}{2}, \frac{n^2+1}{2})$ for odd n and $(n, \frac{n^2}{4} - 1, \frac{n^2}{4} + 1)$ for even n
38	**(a)** 104	$13 \times 360/45$
39	**(c)** 24	Each corner is replaced by 3 new vertices: $8 \times 3 = 24$
40	**(b)** 43	$(29 + 1)(14) - 29 \times 13$

41	**(a)** Both A and R are true and R is the correct explanation of A.	$1296 = 2^4 \times 3^4$, every exponent is even, so it is a perfect square (36^2); R is the test that explains it.
42	**(a)** Both A and R are true and R is the correct explanation of A.	$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$; R is exactly the rule used.
43	**(a)** Both A and R are true and R is the correct explanation of A.	$1 \times 4 + 0 \times 2 + 1 \times 1 = 5$; R is the method used.
44	**(a)** Both A and R are true and R is the correct explanation of A.	Co-interior angles between parallel lines add up to $180°$, which gives A.
45	**(a)** Both A and R are true and R is the correct explanation of A.	$4 + 2 + 3 + 6 = 15$, which is divisible by 3; R is the test used.
46	**(a)** Both A and R are true and R is the correct explanation of A.	Put $a = x$, $b = 5$ in the identity: $x^2 + 2(x)(5) + 25$.
47	**(a)** Both A and R are true and R is the correct explanation of A.	$3 \times 12 = 36 = 4 \times 9$, so the ratios are equal; R is the test.
48	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{25}{100} \times 360 = 90$; R is the rule used.
49	**(a)** Both A and R are true and R is the correct explanation of A.	$6^2 + 8^2 = 36 + 64 = 100 = 10^2$; R is the converse of the theorem.
50	**(a)** Both A and R are true and R is the correct explanation of A.	$300 \div 5 = 60$ per kg, and $60 \times 8 = 480$; R is the property used.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** 5280	Side 73; half other diagonal 55; area $= \frac{1}{2}d_1d_2 = 5280$
2	**(b)** 140	Combine to $a : b : c$ first
3	**(b)** 4	$6(n-2)^2 = 24$ gives $n = 4$
4	**(b)** 12	Sum of first n odd numbers is n^2, so mean $= n$
5	**(b)** 18	$a^2 - b^2 = (a-b)(a+b)$
6	**(c)** 15	$2A/16$
7	**(c)** 3	A k-digit square has $\lceil k/2 \rceil$ digits in its root
8	**(b)** 27	$x = 5$, so 3^3
9	**(a)** MCMXCIV	Break 1994 into thousands, hundreds, tens and units
10	**(c)** 110°	$\angle A + \angle D = 180^\circ$ gives x; base angles $\angle C = \angle D$
11	**(a)** 1	Divisible by 10 and 9
12	**(c)** 25	$p + q = a$, $pq = b$
13	**(d)** 5 : 4	Time is inversely proportional to speed
14	**(d)** 2500	A = P(1+20/100)², CI = A − P
15	**(a)** 17	Foot distances $\sqrt{13^2 - 12^2} = 5$ and $\sqrt{13^2 - 5^2} = 12$ add up to the width
16	**(b)** 24	Remaining food $= (30 - 10) \times 120$ man-days
17	**(c)** 256	$81 \times (4/3)^4$
18	**(c)** 40	Sort the 8 values; median is the mean of the 4th and 5th: 40
19	**(c)** 9	$a^2 - (a-3)(a+3) = 9$
20	**(c)** 840 cm²	Side $= 29$; other half-diagonal $= 21$; area $= \frac{1}{2} \times 40 \times 42$
21	**(a)** -9	$(-9)^3 = -729$
22	**(b)** $\frac{6}{5}$	$\left(\frac{1}{2} + \frac{1}{3}\right)^{-1}$
23	**(c)** 11000_2	Add 1 and carry in base 2
24	**(b)** 6	Perimeter $= 4\times$ side
25	**(b)** 5	Apply the routine and count
26	**(a)** 141	$25^2 - 22^2$
27	**(d)** 32	Adding the denominators keeps the ratio
28	**(a)** 26%	(1+40/100)(1−10/100) − 1
29	**(d)** 36	Side $= 6$ cm, so area $= 6^2 = 36$
30	**(a)** 6	New water $= 36 \times 1/2 = 18$ L
31	**(d)** 4	Side halves every stage: $64/2^4$
32	**(a)** 48	Total distance 240 over total time
33	**(c)** 22	Middle integer is 21
34	**(d)** 1232 m²	$(52 - 8)(36 - 8)$
35	**(a)** 3	7^3 ends in 3
36	**(b)** 0.5×10^6	Convert each to usual form and compare
37	**(b)** 3	The last place has value 3, so the number is a multiple of 3
38	**(a)** isosceles trapezium	Isosceles trapezium has equal diagonals
39	**(b)** 99	9 and 11 have no common factor
40	**(b)** 47	$n^2 - (n-1)^2 = 2n - 1$
41	**(a)** Both A and R are true and R is the correct explanation of A.	A cube has $F = 6$, $V = 8$, $E = 12$ and $6 + 8 - 12 = 2$; R is the general result.

42	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{4+6+8+10+12}{5} = \frac{40}{5} = 8$; R is the definition used.
43	**(a)** Both A and R are true and R is the correct explanation of A.	$2n$ is a multiple of 2 and $2n+1 = 2n+1$ leaves remainder 1; R is the definition behind the expressions.
44	**(a)** Both A and R are true and R is the correct explanation of A.	$12 \times 5 = 60$ cm^2; R is the formula used.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	$(2k)^2 = 4k^2$ is even, so A is true; R is also true ($1+3+5 = 9 = 3^2$) but it does not explain A.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	An odd power of -1 is -1, so A is true; R is also true but is unrelated to the value of $(-1)^{101}$.
47	**(b)** Both A and R are true but R is not the correct explanation of A.	XL = 50 − 10 = 40, so A is true; R is also true but does not explain why XL means 40.
48	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but bisecting does not give equal lengths; equality follows from the right angles of the rectangle.
49	**(b)** Both A and R are true but R is not the correct explanation of A.	The last two digits 24 are divisible by 4, so A is true. R is also a true test but is about 8, so it does not explain A.
50	**(b)** Both A and R are true but R is not the correct explanation of A.	$(a-b)^2 = a^2 - 2ab + b^2 = (b-a)^2$ is true. R is also true but it is a different identity and does not explain A.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 6	$2b + 5 = 17$
2	**(c)** 120°	One part $= 30^\circ$
3	**(d)** 67	LCM plus the common remainder
4	**(b)** $ab = 0$	Compare with $a^2 + 2ab + b^2$
5	**(b)** 27	$x = \frac{9\times12}{4}$
6	**(b)** 50%	SI = 2P so 2P = P×R×T/100 and R = 200/T
7	**(c)** 120	Legs 24 and 45 since $8^2 + 15^2 = 17^2$
8	**(d)** 12	Let ages be $2x$, $5x$: $(2x + 9) : (5x + 9) = 7 : 13$ gives $x = 6$
9	**(b)** 26	$E = 5F/2 = 40$; $V = 2 + E - F$
10	**(b)** 36	$5 \times 24 - 4 \times 21$
11	**(c)** 4	$9(a - b) = 9d$
12	**(c)** 405 cm²	$l + b = 54$, $l = 5b$: $b = 9$, $l = 45$
13	**(d)** 23	Squares from 32^2 to 54^2
14	**(c)** 3^6	$3 \times 3^5 = 3^6$
15	**(c)** 7	Number of binary digits of the largest weight to be measured
16	**(d)** 45°	$x + 3x = 180$
17	**(c)** 59	75 multiples up to 900 minus 16 multiples below 200
18	**(d)** 65	$x^2 + y^2 = (x + y)^2 - 2xy$
19	**(c)** 9 : 12 : 10	Make the middle terms equal using their LCM
20	**(a)** 4%	Loss % = (p²/100) = 20²/100
21	**(b)** 540	Sides 15 and 36; area $= 15 \times 36$
22	**(c)** 2100	One part = ₹300
23	**(d)** 12	$V = 2E/3 = 20$; $F = 2 + E - V$
24	**(c)** 26	Correct sum $= 600 - 40 + 90$
25	**(b)** 14	$(x + 1/x)^2 = x^2 + 2 + 1/x^2$
26	**(b)** 204 cm²	Height $= \sqrt{13^2 - 5^2} = 12$ cm; area $= \frac{1}{2}(12 + 22) \times 12$
27	**(b)** 56	Next square is $49^2 = 2401$
28	**(d)** 5	$3x - 2 = 13$
29	**(b)** 19_{10}	Convert every number to decimal and compare
30	**(b)** 116°	A diagonal bisects the vertex angles; $\angle BCD = 2x$ and $\angle B$ is supplementary to it
31	**(b)** 69	Count multiples from 100 to 999
32	**(d)** 106	$x^2 + y^2 = (x - y)^2 + 2xy$
33	**(a)** 30 years	Let the ages be $5x$ and $3x$; solving $(5x + 6) \times 2 = (3x + 6) \times 3$ gives $x = 6$
34	**(c)** 44100	40000×(1+5/100)²
35	**(d)** 17, 144, 145	Use $(n, \frac{n^2-1}{2}, \frac{n^2+1}{2})$ for odd n and $(n, \frac{n^2}{4} - 1, \frac{n^2}{4} + 1)$ for even n
36	**(d)** 120	$25 \times 360/75$
37	**(d)** 14	6 original faces + 8 new triangular faces
38	**(b)** 37	$(24 + 1)(13) - 24 \times 12$
39	**(c)** 12	$4d + 16$ = sum
40	**(c)** 2000	$2000 \times 125/100 \times 80/100$

41	**(b)** Both A and R are true but R is not the correct explanation of A.	Dividing both terms by 5 gives $3:5$ so A is true; R is also true but does not explain the simplification.
42	**(b)** Both A and R are true but R is not the correct explanation of A.	$800-80=720$, so A is true. R is also true but concerns profit and loss, not the discount, so it does not explain A.
43	**(b)** Both A and R are true but R is not the correct explanation of A.	$25+144=169$ and $9+16=25$, so both are true; but R does not explain A.
44	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true but are about different situations; R does not explain A.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	A triangular prism has 2 triangular and 3 rectangular faces, so A is true; R is true but about pyramids, so it does not explain A.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	The middle value of the five ordered numbers is 9, so A is true; R is also true, but mode is a different measure and does not explain A.
47	**(b)** Both A and R are true but R is not the correct explanation of A.	$3(4)+5=17$ so A is true; R is also true but it does not explain the value.
48	**(b)** Both A and R are true but R is not the correct explanation of A.	A is true by $\frac{1}{2}\times 10\times 8=40$; R is also true but is a separate fact, so it does not explain A.
49	**(c)** A is true but R is false.	$(-3)^3=-27$ so A is true; R is false because cubes of negative integers are negative.
50	**(c)** A is true but R is false.	$3^0+2^0=1+1=2$ so A is true; R is false because $a^0=1$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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